ERFAN RADFAR

Clear Water Bay, Kowloon, Hong Kong SAR · +852 4635 0858
eradfar@connect.ust.hk · erfanradfar200117@gmail.com · github.com/Erfanmode · LinkedIn: erfan-radfar · Google Scholar: Erfan Radfar

RESEARCH INTERESTS

Reinforcement Learning (Credit Assignment, World Models, Causal RL), Self-Supervised & Representation Learning, Robot Learning, Intelligent Control.

EDUCATION

Hong Kong University of Science and Technology (HKUST) **Hong Kong SAR**
Master of Philosophy (M.Phil.) in Electronic and Computer Engineering (GPA: 3.72 / 4.00) Sep 2025 – Aug 2027 (Expected)

- **Thesis Topic:** Contextual Causal Credit Assignment and World-Model Representation Learning in Reinforcement Learning
- **Advisor:** Prof. Wai Ho Mow (Information Theory and Communication Lab)

Sharif University of Technology (SUT) **Tehran, Iran**
Bachelor of Science (B.Sc.) in Mechanical Engineering (GPA: 4.00 / 4.00; 19.10 / 20.00) Sep 2020 – Feb 2025

- **Class Rank:** Ranked 6th among 111 students in the Department of Mechanical Engineering
- **B.Sc. Thesis:** Intelligent Control of a Multi-Finger Robotic Prosthetic Hand for Grasping and Object Manipulation using Deep Reinforcement Learning (Advisor: Prof. A. Taheri)

HONORS & AWARDS

- **1st Rank (Top 0.02%)** in the Iranian National University Entrance Examination for Master's in Mechanical Engineering **2025**
- **Cockrell School of Engineering Graduate Fellowship**, The University of Texas at Austin (awarded for exceptional academic record) **2025**
- **Best Paper Award**, International Conference on Robotics and Mechatronics (ICROM 2024) **2024**
- **Gold Medal (1st Rank)**, National Olympiad of Mechanical Engineering of Iranian Universities **2023 – 2024**
- **1st Rank Student Award in B.Sc.** (among 111 students, two consecutive academic years: 2020–2021 & 2021–2022), Sharif University of Technology **2020 – 2022**
- **Silver Medal**, National Olympiad of Physics for High School Students, Iran **2019**

PUBLICATIONS

- Arshia Rezaei, **Erfan Radfar**, Amin Talaeizadeh, and Aria Alasty, "**Physics-Informed Deep Learning-Based Monocular Vision & IMU Fusion for Extracting Position & Heading Data of Multirotor**," *International Conference on Robotics and Mechatronics (ICROM)*, 2024. [**Best Paper Award**]
- **Erfan Radfar**, "**Stock Market Trend Prediction using Deep Neural Network via Chart Analysis: A Practical Method or a Myth?**," *Humanities and Social Sciences Communications (Springer Nature)*, 2025.

RESEARCH EXPERIENCE

Information Theory and Communication Lab, HKUST **Hong Kong SAR**
Graduate Research Assistant (Advisor: Prof. Wai Ho Mow) *Fall 2025 – Present*

- **Contextual Causal Credit Assignment in RL (M.Phil. Thesis):** Formulating a context-aware causal action-value attribution framework generalizing Compositional Credit Assignment (COCOA) beyond temporal proximity. Establishing theoretical properties and empirical scalability of action-contribution baselines across discrete and continuous control benchmarks.
- **World Models & Latent Hindsight Representations:** Investigated Counterfactual Credit Assignment (CCA) coupled with forward imagination world models (Dreamer and Transformer-based TWISTER with contrastive predictive coding) and JEPA-style self-supervised sequence representations; systematically analyzed identifiability and scalability bottlenecks of hindsight latent CCA in complex dynamical systems.
- **Physical-Layer Signal Authentication (Funded Industrial RA Project):** Developed machine learning pipelines (SVM-based classification with Gaussian channel coding and demodulation) and an interactive verification GUI for physical-layer hardware impairment authentication under spoofing attacks.

Social and Cognitive Robotics Laboratory, Sharif University of Technology **Tehran, Iran**
Undergraduate Research Assistant (Advisor: Prof. A. Taheri) *Summer 2024 – Summer 2025*

- **Vision-Based Dexterous Manipulation via Deep RL:** Trained Proximal Policy Optimization (PPO) policies integrated with RGB-D visual perception (object detection, segmentation, and stereo depth estimation) for dexterous grasping and manipulation on a high-DoF anthropomorphic prosthetic hand.
- **Generalization Performance:** Achieved an **88.4% grasp success rate** on abstract training geometries and **69.0% zero-shot generalization** on unseen household objects, matching state-of-the-art proprioceptive baselines.
- **Learning-Augmented Optimal Control:** Integrated Deep Reinforcement Learning with Model Predictive Control (MPC) and performed nonlinear system identification, trajectory planning, and parameter fine-tuning for multi-link robotic systems.

SELECTED TECHNICAL PROJECTS

3D Scene Reconstruction & Generative Vision Pipelines **Fall 2025**
Course Project Series (ELEC 5630), HKUST (Instructor: Prof. Ping Tan) *Hong Kong SAR*

- Implemented end-to-end 2D/3D computer vision pipelines including a 3D point cloud diffusion model, Neural Radiance Fields (NeRF), 3D Gaussian Splatting, and multi-view Structure-from-Motion (SfM) reconstruction.

Multirotor Design, Dynamical Estimation & Obstacle-Aware Path Planning **Summer 2024**
Aerial Robotics Project, SUT (Instructor: Dr. A. Talaeizadeh) *Tehran, Iran*

- Designed cascaded PID flight control architectures integrated with Kalman filter dynamical state estimation and real-time obstacle avoidance trajectory generation in MATLAB/Simulink.

Robust Trajectory Control & Motion Planning for an RRP Manipulator **Fall 2023**
Robotics Project, SUT (Instructor: Prof. S. Behzadipour) *Tehran, Iran*

- Implemented graph-based dynamic programming planners (Dijkstra and A*) and artificial potential fields alongside inverse-dynamics robust control under external disturbances and actuator jitter.

INDUSTRY EXPERIENCE

Center of Excellence in Design, Robotics and Automation (CEDRA) **Tehran, Iran**
Robotics & Autonomous Systems Intern *Jun 2023 – Aug 2023*

- Developed multi-sensor perception fusion algorithms using Extended Kalman Filtering (EKF) with neural-network-assisted filter tuning, and built state-space autonomous vehicle navigation simulations in Python and MATLAB.

TEACHING EXPERIENCE

- **Graduate Teaching Assistant**, Department of Electronic and Computer Engineering, HKUST **Spring 2026**
  Course: Probability and Random Processes (Instructor: Prof. W. H. Mow)
- **Undergraduate Teaching Assistant (5 Appointments)**, Department of Mechanical Engineering, SUT **Spring 2023 – Fall 2024**
  Courses: Robotics (Prof. S. Behzadipour, Fall 2024); Dynamics (Dr. S. Mohammadi, Fall 2024 & Dr. A. Taheri, Fall 2023); Numerical Computation (Dr. S. Mohammadi, Spring 2024); Industrial Design & Drawing (Dr. H. Haghshenas, Spring 2023)
- **Physics Olympiad Instructor & Exam Problem Designer**, Elite High Schools of Karaj, Iran **Fall 2019 – Spring 2024**
  Trained students and designed national-level problem sets for the Iranian National Physics Olympiad.

RELEVANT COURSEWORK

- **Graduate (HKUST):** Convex Optimization (ELEC 5470), Stochastic Optimization for Wireless Systems & Federated Learning (ELEC 5460), Computer Vision (ELEC 5630), Networked Sensing, Estimation & Control (ELEC 5650), Digital Communications & Wireless Systems (ELEC 4110).
- **Undergraduate (SUT):** Intelligent Systems (Deep Learning & Genetic Algorithms), Robotics & Path Planning, Aerial Robotics, Automatic Control, Fundamentals of Controller Design, Dynamics of Machinery, Mechanical Vibrations, Numerical Computations, Programming in C.

TECHNICAL SKILLS & LANGUAGES

- **Programming & Machine Learning:** Python, PyTorch, C, C++, MATLAB, Simulink, LaTeX, Ubuntu Linux
- **Robotics, Simulation & CAD:** MuJoCo, ROS 2, Gazebo, SolidWorks, CATIA, Abaqus, Arduino, PLC
- **Languages:** English: Fluent (C2 Proficiency — **IELTS Academic Overall 8.5**: Listening 9.0, Reading 9.0, Writing 7.5, Speaking 7.5; Oct 2024) · Persian: Native